Bang-Shien Chen

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Education

PhD. in Electrical and Computer Engineering *Sep 2025 - Present*

Northeastern University
GPA: 4.0/4.0
Research Advisor: Prof. David M. Rosen

M.Sc. in Mathematics *Sep 2023 - Jun 2025*

National Taiwan Normal University
GPA: 4.3/4.3
Research Advisor: Prof. Jann-Long Chern
Co-Advisor: Prof. Chih-Wei Huang

B.Sc. in Mathematics *Sep 2018 - Jun 2022*

National Taiwan Normal University
GPA: 3.18/4.3

Academic Experience

Research Visitor hosted by Prof. Shuzhong Zhang, University of Minnesota. *Jun 2024 - Aug 2024*

Teaching Assistant of Linear Algebra by Prof. Chu-Chin Hu. *Sep 2023 - Jun 2024*

Research Assistant of Prof. Jann-Long Chern. *Jul 2022 - Jan 2023, May 2023 - Aug 2023*

Awards & Achievements

Draper Scholar | *link* *Sep 2025 - Present*
A scholarship designed to transform exceptional students into innovative professionals by integrating thesis research with hands-on experience.

NCTS Research Abroad Program | *link* *Jul 2024 - Aug 2024*
A research abroad program financially supported by the Mathematics Division of National Center of Theoretical Sciences, worked with Prof. Shuzhong Zhang at the University of Minnesota.

Taiwan-Japan Joint Workshop, Presentation Award | *link* *Feb 2024*
Out of 40 speakers in the 14th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics.

IBM Qiskit Hackathon Taiwan, 1st Prize | *link* *Jul 2022*
Out of 12 teams in the final round of the Hackathon.

Taiwan-Japan Joint Workshop, Presentation Award | *link* *Feb 2022*
Out of 42 speakers in the 12th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics.

Presentation & Publication

Presentations

1. Workshop on Advances in Continuous Optimization 2024, RIKEN Center for Advanced Intelligence Project, Tokyo, Japan, November 25-26, 2024.
2. 2024 NCTS Student Workshop on Scientific Computing, National Taiwan University, Taipei, Taiwan, May 6, 2024.
3. 14th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics, Meiji University, Tokyo, Japan, February 27-29, 2024.
4. Workshop on Advances in Continuous Optimization 2023, The University of Tokyo, Tokyo, Japan, July 24-25, 2023.
5. 2022 Summer school on Quantum Computer, Asia University, Taipei, Taiwan, August 21 - September 24, 2022.
6. Seminar on Quantum Theory and Computation, National Taiwan Normal University, Taipei, Taiwan, April 26 - May 24, 2022.
7. 12th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics, (virtual), February 28 - March 1, 2022.

Publications

[1] **B.S. Chen**, C.E. Eaton, D.M. Rosen. (2026). *PnP*: A certifiable solver for Perspective-n-Point.* arXiv:2610.31797.

[2] U. Rai, Z. Xu, **B.S. Chen**, D.M. Rosen. (2026). *A QCQP-Representable IMU Pre-Integration Factor for Certifiable State Estimation.* arXiv:2609.38048.

[3] Y.J. Li, Y.K. Lin, **B.S. Chen**, C.W. Huang, J.L. Chern, C.C. Sun. (2026). *Open-Vocabulary Semantic Segmentation for Dynamic 3D Scenes Using Scene Flow Estimation.* IEEE Robotics and Automation Letters, 11(7), 8140-8147, doi: 10.1109/LRA.2026.3693993.

[4] **B.S. Chen**, Y.K. Lin, J.Y. Chen, C.W. Huang, J.L. Chern, C.C. Sun. (2024). *FracGM: A Fast Fractional Programming Technique for Geman-McClure Robust Estimator.* IEEE Robotics and Automation Letters, 9(12), 11666-11673, doi: 10.1109/LRA.2024.3495372.

[5] **B.S. Chen**, J.L. Chern. (2022). *Generating quantum feature maps for SVM classifier.* arXiv:2207.11449.

Skills

Research: Optimization, Robotics, Computer Vision, Machine Learning

Programming: Python, C++, Matlab

Languages: TOEFL: 109, GRE: 320